

COMPUTER SCIENCE**PAPER 1 (THEORY)****Maximum Marks: 70****Time Allowed: Three Hours***Reading Time: Additional Fifteen Minutes***Instructions to Candidates**

1. You are allowed an additional fifteen minutes for only reading the question paper.
2. You must NOT start writing during the reading time.
3. This question paper has 12 printed pages.
4. It is divided into two parts: Part I and Part II.
5. It has eleven questions in all.
6. Part I is compulsory and has two questions.
7. While attempting Multiple Choice Questions in Part I, you are required to write only ONE option as the answer.
8. Part II is divided into three sections: A, B and C.
9. Each section in Part II has three questions. Any two questions have to be attempted from each section.
10. The intended marks for questions are given in brackets [].

Instruction to Supervising Examiner

1. Kindly read aloud the Instructions given above to all the candidates present in the examination hall.

PART I – 20 MARKS*Answer all questions.**While answering questions in this Part, briefly indicate your working and reasoning, wherever required.***Question 1**

- (i) The complement of the Boolean expression $(A + B') \cdot (A' + C)$ is: (*Application*) [1]
- (a) $A' \cdot B + A \cdot C'$
 - (b) $A \cdot B' + A' \cdot C$
 - (c) $(A' + B) \cdot (A + C')$
 - (d) $A' \cdot B \cdot A \cdot C'$

- (ii) Study the given propositions and the statements marked Assertion and Reason that follow. [1]
Choose the correct option on the basis of your analysis.

p = A quadrilateral is a square

q = The quadrilateral is a rectangle

$s1 = p \Rightarrow q$

$s2 = \sim p \Rightarrow \sim q$

Assertion: $s2$ is the inverse of $s1$.

Reason: The inverse of a true conditional statement is always true.

(Analysis)

- (a) Both Assertion and Reason are true and Reason is the correct explanation for Assertion.
- (b) Both Assertion and Reason are true but Reason is not the correct explanation for Assertion.
- (c) Assertion is true and Reason is false.
- (d) Both Assertion and Reason are false.
- (iii) According to the Principle of Duality, the Boolean equation $P' + P \cdot Q + 0 = P' + Q$ will be equivalent to: (Application) [1]
- (a) $P' \cdot (P + Q) \cdot 1 = P' \cdot Q$
- (b) $P \cdot (P' + Q') \cdot 1 = P \cdot Q'$
- (c) $P' \cdot (P + Q) \cdot 0 = P' \cdot Q$
- (d) $P + (P' \cdot Q') + 1 = P + Q'$
- (iv) The Boolean equations $A + A = A$ and $A \cdot A = A$ correspond to: (Recall) [1]
- (a) Idempotent law
- (b) Law of identity
- (c) Absorption law
- (d) Involution law
- (v) The canonical POS form of $F(X, Y, Z) = \pi(0, 5, 6)$ is: (Application) [1]
- (a) $(X + Y + Z) \cdot (X' + Y + Z') \cdot (X' + Y' + Z)$
- (b) $(X' + Y' + Z') \cdot (X + Y' + Z) \cdot (X + Y + Z')$
- (c) $X' \cdot Y' \cdot Z' + X \cdot Y' \cdot Z + X \cdot Y \cdot Z'$
- (d) $(X + Y + Z) \cdot (X' + Y + Z') \cdot (X' + Y' + Z')$
- (vi) The worst case complexity of the following code segment is: [1]

```
for (int i = 1; i <= n; i++)  
    for (int j = i; j <= n; j++)  
        count++;
```

(Analysis)

- (a) $O(n)$
- (b) $O(n^2)$
- (c) $O(n \log n)$
- (d) $O(2n)$

(vii) Consider the following class:

[1]

```
class Book
{
    String title;
    Book()
    {
        title = "none";
    }
    Book(String t)
    {
        title = t;
    }
}
```

Which of the following statements are valid?

- I. A constructor has the same name as its class and no return type.
- II. A class may have more than one constructor, as shown above.
- III. A method **void Book()** added to this class would be a third constructor.

(Application)

- (a) I and II
- (b) II and III
- (c) I and III
- (d) Only I

(viii) Given below are two statements marked Assertion and Reason. Read the two statements carefully and choose the correct option.

[1]

Assertion: An abstract class can have a constructor.

Reason: The constructor of an abstract class is executed when an object of its sub class is created.

(Analysis)

- (a) Both Assertion and Reason are true and Reason is the correct explanation for Assertion.
- (b) Both Assertion and Reason are true but Reason is not the correct explanation for Assertion.
- (c) Assertion is true and Reason is false.
- (d) Both Assertion and Reason are false.

(ix) Write the contrapositive of the statement: "If a student scores above 90%, then the student gets a scholarship." *(Application)*

[1]

(x) Why are the rows and columns of a Karnaugh map labelled in the order 00, 01, 11, 10 and not 00, 01, 10, 11? *(Understanding)*

[1]

Question 2

(i) Convert the following infix notation to postfix form:

[2]

$A * (B + C / D) - E / (F - G)$ *(Application)*

(ii) A matrix $M[-5 \dots 8, 3 \dots 12]$ is stored in the memory with each element requiring 2 bytes of storage. If the base address is 1850, find the address of $M[4][9]$ when the matrix is stored in column major wise. *(Application)*

[2]

- (iii) The following function `count( )` is a part of some class. Assume `n` and `k` are positive integers. Answer the questions given below along with dry run / working.

```
int count(int n, int k)
{
    if (n == 0)
        return 0;
    else if (n % 10 > k)
        return 1 + count(n / 10, k);
    else
        return count(n / 10, k);
}
```

- (a) What will the function `count( )` return when `count(739412, 3)` is called? (*Analysis*) [2]
- (b) What is the function `count( )` performing apart from recursion? (*Analysis*) [1]
- (iv) The following function `isHarshad( )` is a part of some class. It returns true if the positive integer `n` is a Harshad number, i.e. `n` is divisible by the sum of its digits. [Example: 156 is a Harshad number, as $1 + 5 + 6 = 12$ and 156 is divisible by 12.]

There are some places in the code marked by ?1?, ?2?, ?3? which may be replaced by a statement/expression so that the function works properly.

```
boolean isHarshad(int n)
{
    int sum = ?1?;
    for (int t = n; t > 0; ?2?)
        sum = sum + t % 10;
    return ?3?;
}
```

- (a) What is the expression or statement at ?1? (*Analysis*) [1]
- (b) What is the expression or statement at ?2? (*Analysis*) [1]
- (c) What is the expression or statement at ?3? (*Analysis*) [1]

PART II – 50 MARKS

Answer six questions in this part, choosing two questions from Section A, two from Section B and two from Section C.

SECTION A

Answer any two questions.

Question 3

- (i) A state sports academy selects a player for the state team (S) if any one of the following conditions is satisfied: [5]

- The player is physically fit and has played at least three national events.

OR

- The player is physically fit, is under 19 years of age and is recommended by the coach.

OR

- The player has played at least three national events and is recommended by the coach but is not physically fit (selected as a reserve).

The inputs are:

INPUTS

- P The player is physically fit.
N The player has played at least three national events.
C The player is recommended by the coach.
U The player is under 19 years of age.

(In all the above cases, 1 indicates yes and 0 indicates no.)

Output: S – denotes that the player is selected [1 indicates yes and 0 indicates no in all cases].

Draw the truth table for the inputs and outputs given above and write the SOP expression for $S(P, N, C, U)$.

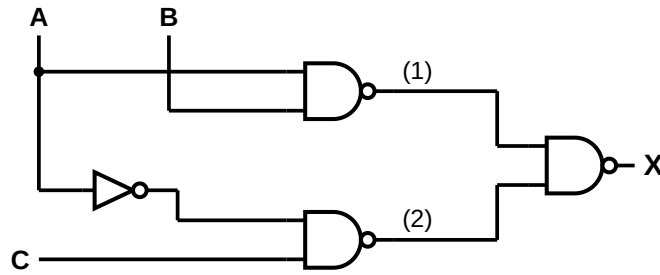
(Application)

- (ii) Reduce the above expression $S(P, N, C, U)$ by using a 4-variable Karnaugh map, showing the various groups (i.e. octal, quads and pairs). Draw the logic gate diagram for the reduced expression using NAND gates only. Assume that the variables and their complements are available as inputs. (Create) [5]

Question 4

- (i)
- (a) Reduce the Boolean function $F(P, Q, R, S) = \pi(0, 1, 4, 5, 6, 7, 8, 9, 14)$ by using a 4-variable Karnaugh map, showing the various groups (i.e. octal, quads and pairs). [4]
(Application)
- (b) Draw the logic gate diagram for the reduced expression. Assume that the variables and their complements are available as inputs. (Create) [1]

- (ii) From the logic diagram given below, derive the Boolean expressions at (1), (2) and X. Name the combinational circuit that the diagram represents. (*Application*) [3]



- (iii) Using a truth table, state whether the following proposition is a tautology, contradiction or contingency: $(p \wedge \sim q) \wedge (\sim p \vee q)$ (*Application*) [2]

Question 5

- (i)
- (a) What is a decoder? (*Recall*) [1]
 - (b) Draw the truth table and the logic circuit of a 3 to 8 (octal) decoder. (*Application*) [3]
 - (c) Which multiplexer can be derived from an octal (3 to 8) decoder? State how. (*Understanding*) [1]
- (ii) Using the laws of Boolean algebra (propositional logic), prove that:
 $(a \Rightarrow b) \wedge (a \Rightarrow c) = a \Rightarrow (b \wedge c)$ (*Application*) [3]
- (iii) Write the minterm and the maxterm when the inputs are $A = 1$, $B = 0$, $C = 1$ and $D = 1$. (*Application*) [2]

SECTION B

Answer any two questions.

Each program should be written in such a way that it clearly depicts the logic of the problem.

This can be achieved by using mnemonic names and comments in the program.

(Flowcharts and Algorithms are not required.)

The programs must be written in Java.

Question 6

A **Dudeney number** is a positive integer that is a perfect cube such that the sum of its digits is equal to the cube root of the number. [10]

Example: $5832 = 18^3$ and $5 + 8 + 3 + 2 = 18$; $4913 = 17^3$ and $4 + 9 + 1 + 3 = 17$. $1000 = 10^3$ is not a Dudeney number, as $1 + 0 + 0 + 0 = 1$.

A class Dudeney has been defined to check whether a number is a Dudeney number. Some of the members of the class are given below:

Class name : **Dudeney**

Data members/instance variables:

num : to store a positive integer

Methods/Member functions:

Dudeney() : default constructor to initialise the data member with a legal initial value

void input() : to accept the number num

int sumDigits(int x) : returns the sum of the digits of x using the **recursive technique**

int cubeRoot(int n) : returns the cube root of n if n is a perfect cube, otherwise returns -1

void isDude() : checks whether num is a Dudeney number by invoking the functions cubeRoot() and sumDigits(), and displays an appropriate message

Specify the class Dudeney giving details of the constructor(), void input(), int sumDigits(int), int cubeRoot(int) and void isDude(). Define a main() function to create an object and call the functions accordingly to enable the task.

(Create)

Question 7

Design a class MatRev to reverse each element of a matrix.

[10]

Example:

31	408	7
125	60	19

becomes

13	804	7
521	6	91

Some of the members of the class are given below:

Class name : **MatRev**

Data members/instance variables:

arr[][] : to store integer elements
m : to store the number of rows
n : to store the number of columns

Methods/Member functions:

MatRev(int mm, int nn) : parameterised constructor to initialise the data members m = mm and n = nn
void fillarray() : to enter the elements of the matrix
int reverse(int x) : returns the reverse of the number x
void revMat(MatRev P) : reverses each element of the array of the parameterised object and stores it in the array of the current object
void show() : displays the array elements in matrix form

Define the class MatRev giving details of the constructor(), void fillarray(), int reverse(int), void revMat(MatRev) and void show(). Define the main() function to create objects and call the functions accordingly to enable the task.

(Create)

Question 8

Given two strings, an original string and a mask string, remove from the original string every character that occurs in the mask string. [10]

Example 1: ORIGINAL STRING : communication
 MASK STRING : mont
 OUTPUT : cuicai

Example 2: ORIGINAL STRING : information
 MASK STRING : form
 OUTPUT : inatin

A class StringOp is defined as follows to perform the above operation. Some of the members of the class are given below:

Class name : **StringOp**

Data members/instance variables:

str : to store the original string
msk : to store the mask string
nstr : to store the resultant string

Methods/Member functions:

StringOp() : default constructor to initialise the data members with legal initial values
void accept() : to accept the original string str and the mask string msk in lower case
void form() : to form the new string nstr after removing from str every character present in msk
void display() : to display the original string, the mask string and the newly formed string nstr

Specify the class StringOp giving details of the constructor(), void accept(), void form() and void display(). Define a main() function to create an object and call all the functions accordingly to enable the task.

(Create)

SECTION C

Answer any two questions.

Each program should be written in such a way that it clearly depicts the logic of the problem stepwise.

This can be achieved by using comments in the program and mnemonic names or pseudo codes for algorithms. The programs must be written in Java and the algorithms must be written in general / standard form, wherever required / specified.

(Flowcharts are not required.)

Question 9

- (i) A book rack in a library can hold at the most 50 books, each identified by an integer accession number. A book can be placed on, or taken off, the rack at either end — the front or the rear. [4]

Define a class BookRack with the following details:

Class name : **BookRack**

Data members/instance variables:

bk[] : array to hold the accession numbers of the books
cap : stores the maximum capacity of the rack
front : to point to the index of the front end
rear : to point to the index of the rear end

Methods/Member functions:

BookRack(int max) : constructor to initialise cap = max, front = rear = -1 and to create the integer array
void addFront(int v) : to add a book at the front end if possible, otherwise display the message "RACK FULL AT FRONT"
void addRear(int v) : to add a book at the rear end if possible, otherwise display the message "RACK FULL AT REAR"
int removeFront() : to remove and return the accession number at the front end; if the rack is empty, return -9999
int removeRear() : to remove and return the accession number at the rear end; if the rack is empty, return -9999

Specify the class BookRack giving details of the functions void addRear(int) and int removeFront(). Assume that the other functions have been defined.

The main() function and algorithm need NOT be written.

(Create)

- (ii) Name the entity described above and state its principle. *(Understanding)* [1]

Question 10

A courier company charges for delivering a parcel as per the table given below. For express delivery a surcharge of 25% of the charge is added.

[5]

Weight of the parcel	Charge
Up to 1 kg	Rs 50 (flat)
Above 1 kg and up to 5 kg	Rs 50 + Rs 30 per kg above 1 kg
Above 5 kg	Rs 170 + Rs 20 per kg above 5 kg

A super class Parcel has been defined to store the details of a parcel. Define a sub class Courier to calculate the delivery charge. The details of the members of both the classes are given below:

Class name : **Parcel**

Data members/instance variables:

pid : to store the parcel id

wt : to store the weight of the parcel in kg (in decimals)

Methods/Member functions:

Parcel(...) : parameterised constructor to assign values to the data members

void show() : to display the parcel details

Class name : **Courier**

Data members/instance variables:

express : to store true for express delivery, otherwise false

charge : to store the delivery charge (in decimals)

Methods/Member functions:

Courier(...) : parameterised constructor to assign values to the data members of both the classes

void calCharge() : to calculate the delivery charge as per the table given above, including the express surcharge

void show() : to display the parcel details along with the delivery type and the charge

Assume that the super class Parcel has been defined. Using the concept of inheritance, specify the class Courier giving the details of the constructor(...), void calCharge() and void show(). The super class, main function and algorithm need NOT be written.

(Create)

Question 11

- (i) A linked list is formed from the objects of the class Node. The class structure of the Node is given below: [2]

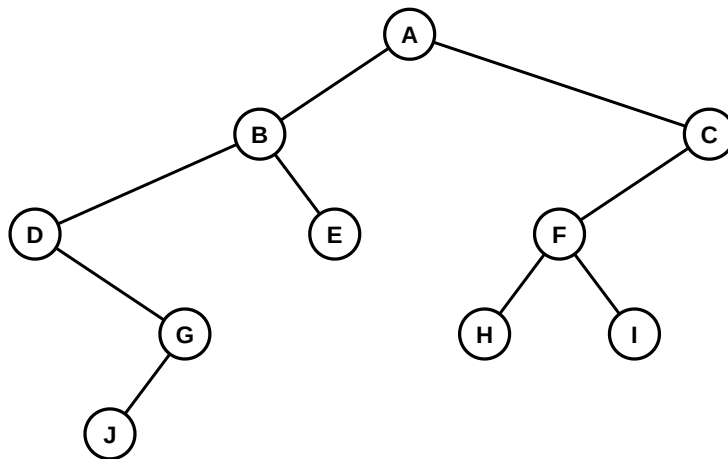
```
class Node
{
    int data;
    Node next;
}
```

Write an Algorithm OR a Method to search for the value v in an existing linked list and return its position (the first node is at position 1), or -1 if v is not present. The method declaration is as follows:

`int position(Node start, int v)`

(Analysis)

- (ii) Answer the following questions from the diagram of a Binary Tree given below:



- (a) Write the postorder traversal of the left subtree. (*Understanding*) [1]
- (b) State the level of node H and the height of the tree (the root is at level 0). (*Understanding*) [1]
- (c) Name the internal nodes of the right subtree. (*Understanding*) [1]

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